

Appendix Y. Employment Projection and Occupational Forecast Methods

This appendix documents the workflow used to derive the SAM-standard employment projections and occupation metrics that populate the updated dashboard. It references the repository structure for reproducibility and describes the analytical intent behind each processing stage.

Overview

The SAM-based employment pipeline combines three main inputs:

1. **Automotive attribution shares** from the Michigan Social Accounting Matrix (SAM) processed via `scripts/build_sam_auto_shares_v2.py` and stored in `data/intermediate/sam_naics_shares_v2/sam_auto_naics4_mobility38.csv`.
2. **Growth rates** for multiple projection scenarios pulled from `data/raw/naics-level-employment-projections.csv` and normalized by `scripts/build_employment_projection_comparison.py`, producing `data/intermediate/employment_projection_comparison.csv`.
3. **Segment-to-occupation staffing shares** drawn from MCDA/Lightcast files in `data/processed/mcda_staffing_detailed_2021_2024.csv`.
4. **BLS occupational drift factors** derived from Employment Projections Table 1.9 detail files, aggregated to segments in `data/processed/us_staffing_segments_summary.csv`.

The script `scripts/build_sam_standard_dashboard_data.py` orchestrates the full process. It applies SAM-derived shares to baseline employment, projects forward under several growth scenarios, and distributes segment totals to occupations. All outputs are written to `data/processed/sam_auto_dashboard/` for consumption by updated dashboards and notebooks.

Step-by-Step Workflow

1. Load SAM Shares and Projection Rates

- File: `data/intermediate/sam_naics_shares_v2/sam_auto_naics4_mobility38.csv`
- Purpose: provides NAICS-level `auto_share_of_output` along with the mobility segment and stage assignments. Only upstream industries (stage = "Upstream") plus NAICS 5413, 5414, and 5417 are adjusted; all other NAICS retain their full 2024 QCEW employment.
- File: `data/intermediate/employment_projection_comparison.csv`
- Purpose: supplies Moody's Michigan/US, DTMB, and BLS employment change rates for each NAICS industry.

The merged dataset is the basis for all subsequent calculations. For each NAICS, the script stores:

```
auto_share_applied = auto_share_of_output if stage == "Upstream"
                    or naics_code in {5413, 5414, 5417}
                    else 1.0
```

```
auto_base_employment = employment_qcew_2024 * auto_share_applied
```

2. Build NAICS Time Series

Using the annualized variants of the six-year growth rates, the script creates a time series from 2024 through 2034 for each projection method (Moody’s MI, Moody’s US, DTMB MI, BLS US). All four scenarios share the same SAM-derived baseline; they diverge only due to projection rates.

Output: `data/processed/sam_auto_dashboard/sam_employment_naics_timeseries.csv` with distinct columns for raw employment, auto-adjusted employment, rate/cagr metadata, and the applied share.

Analytical intent: track how much of each industry’s employment is treated as automotive, and allow downstream comparisons across projection scenarios.

3. Aggregate to Segments and Stages

The NAICS time series roll up to segment- and stage-level files:

- `sam_employment_segment_timeseries.csv` – totals by segment and projection method. Includes `adjustment_source=sam_mi` and the calculated auto share ratio (auto employment ÷ raw employment).
- `sam_employment_stage_timeseries.csv` – similar aggregation by production stage, plus an `Upstream+Core` combined row.

These tables represent the input panels for segment/stage visualizations.

4. Prepare Occupation Forecast Inputs

The script loads detailed staffing shares from `data/processed/mcda_staffing_detailed_2021_2024.csv` and retains the 2024 proportional distribution of occupations within each segment. The BLS staffing table provides 2024 and 2034 shares for each occupation within every mobility segment. For each occupation-segment pair the script computes an annualised drift factor, applies it to the Michigan baseline share, and normalises the resulting shares so they continue to sum to one within each segment-year. Segment employment totals from each projection scenario are then multiplied by the drift-adjusted shares to obtain occupation-level employment estimates. Baseline openings from the MCDA data are scaled in proportion to employment changes so that occupation-level openings track the projected job counts.

Outputs:

File	Description
<code>sam_occ_segment_totals_2024_2034.csv</code>	Complete SOC-level panel by segment, method, and year.
<code>sam_occ_segment_totals_2030.csv</code>	Snapshot for 2030 only (useful for quick reporting).
<code>sam_occ_segment_totals_validation.csv</code>	Check that segment-level sums match <code>sam_employment_segment_timeseries.csv</code> .

5. Summary of Repository Artifacts

Processing scripts (under `scripts/`):

- `build_sam_auto_shares_v2.py` – generate SAM attribution shares from raw Michigan SAM data.
- `build_employment_projection_comparison.py` – normalize the raw growth rate file.
- `build_sam_standard_dashboard_data.py` – the integrated workflow that creates SAM-adjusted projections and occupation outputs.

Key intermediate data (under `data/intermediate/`):

- `sam_naics_shares_v2/sam_auto_naics4_mobility38.csv` – NAICS-level SAM shares.
- `employment_projection_comparison.csv` – cleaned projection rates.

Processed deliverables (under `data/processed/sam_auto_dashboard/`):

- `sam_employment_naics_timeseries.csv`
- `sam_employment_segment_timeseries.csv`
- `sam_employment_stage_timeseries.csv`
- `sam_segment_totals_for_occ.csv` (segment totals in the format required by other tooling)
- `sam_occ_segment_totals_2024_2034.csv`
- `sam_occ_segment_totals_2030.csv`
- `sam_occ_segment_totals_validation.csv`

Analytical Purpose

The SAM-standard workflow formalizes how much of each industry's employment is considered part of the automotive supply chain. By restricting the adjustment to upstream industries and specific OEM support NAICS (5413, 5414, 5417), the method isolates the portion of employment that most plausibly scales with vehicle production and supplier demand. Segment and stage aggregations allow stakeholders to compare scenarios across the mobility value chain, while the occupation outputs translate segment-level swings into potential workforce impacts.

These files feed the new dashboard revision, enabling consistent comparisons across projection sources while anchoring the analysis in the Michigan SAM as the chosen standard for industry attribution.